RODRIGO GOMES DE ARAÚJO

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Education

Master's — **Sep 2025 – Present**
Computer Science — *University of Porto (FCUP)*

Bachelor's — **Sep 2022 – Jul 2025**
Computer Science and Engineering — *University of Porto (FEUP)*

Experience

Murph | *Co-founder & Lead Engineer* | *Rust, gitoxide, gRPC, Ed25519, IPLD* — **Jul 2026 – Present**

- Co-founded a **3-person startup** (pre-launch) and wrote most of the codebase.
- Built a platform where **AI agents and humans** ship changes together, merging version control, tasks and CI.
- Encoded the merge pipeline as **Rust typestates**, so no change reaches main without verification and review.
- Designed **Ed25519-signed approvals**, tiered authority for agents and humans and a tamper-evident history.
- Built a Git layer with **compare-and-swap** ref updates, so agents and verification run in parallel worktrees.
- Implemented **crash-safe persistence** that recovers state from Git via atomic checkpoints.
- Served reads through a **byte-weighted LRU cache** of content-addressed Git objects, needing no invalidation.

Euronext | *AI DevOps Trainee* | *Group Data Delivery* — **Nov 2026 – Present**

Research

CaST (Cryptographically-aware Session Types) | *Master's Thesis* — **Jul 2026 – Present**

- Extending session types to track **cryptographic resource usage**, rejecting protocol misuse at compile-time.
- Formalizing the extension with a **soundness proof** and integrating it into the FreeST compiler (Haskell).
- Built a **Kademlia DHT in FreeST** and a syntax highlighting extension for the Zed editor.

Projects

PoW Blockchain for Decentralized Auctions | *Data Systems Security*, 19/20 — *Rust, Tokio, libp2p*

- Surveyed Bitcoin, Ethereum, Solana and **churn and attack research** to guide network and consensus design.
- Built a peer-to-peer overlay on a **Kademlia DHT** (as in Ethereum's discovery layer) for peer discovery.
- Designed a **Proof-of-Work blockchain** with BLAKE2b, Ed25519, Merkle trees and longest-chain consensus.
- Created an auction client that uses the chain as a public ledger for recording and verifying bids.
- Simulated bootstrap, honest and malicious nodes in Docker to study adversarial behaviour.

Distributed Protocols | *Distributed Systems*, 19/20 — *Rust, Tokio, Tonic*

- **Totally ordered multicast** with Lamport clocks, hardened against reused, backdated and far-future timestamps.
- **Token-ring mutual exclusion** with failure detection and automatic ring healing.
- **Push-pull gossip** with exponential anti-entropy for state convergence.

Real-Time Distributed Synthesizer | *Embedded Systems*, 19/20 — *C, miniaudio, Arduino, Raspberry Pi*

- Built a **real-time synthesizer** split across an Arduino controller, a Raspberry Pi engine and an Android client.
- Wrote a **lock-free audio engine** in C with atomics and ring buffers, applying control events every audio callback.
- Designed a **node-based processing chain** with polyphony, 4 waveforms, ADSR, LFO and low/high-pass filters.
- Sent controller state as **checksummed serial packets** and streamed PCM frames to Android over **UDP**.

Achievements and Awards

Top 10 of 130 Startups, Fall 2026 Academia Cohort with Murph | Unicorn Factory Lisboa — **Oct 2026**

2nd Place, IEEE RetroJam 2025 (48-hour game in Rust) | IEEE UP Student Branch — **Oct 2025**

1st Place in the National Robotics Championship, RoboCup 2016 Participant | UPB — **May 2016**

Technical Skills

Languages: Rust, C/C++, Haskell, OCaml, Java, Python, SQL

Distributed Systems & Cryptography: Tokio, libp2p, Kademlia, gRPC, gitoxide, Ed25519, BLAKE2b, Merkle Trees

Data & Infrastructure: PostgreSQL, PostGIS, FastAPI, Docker, Terraform, Git, UNIX/Linux

Web: React, Dioxus, Bonsai, Dream, HTML5, CSS

Spoken Languages: Portuguese (Native), English (Fluent), Spanish (Intermediate)